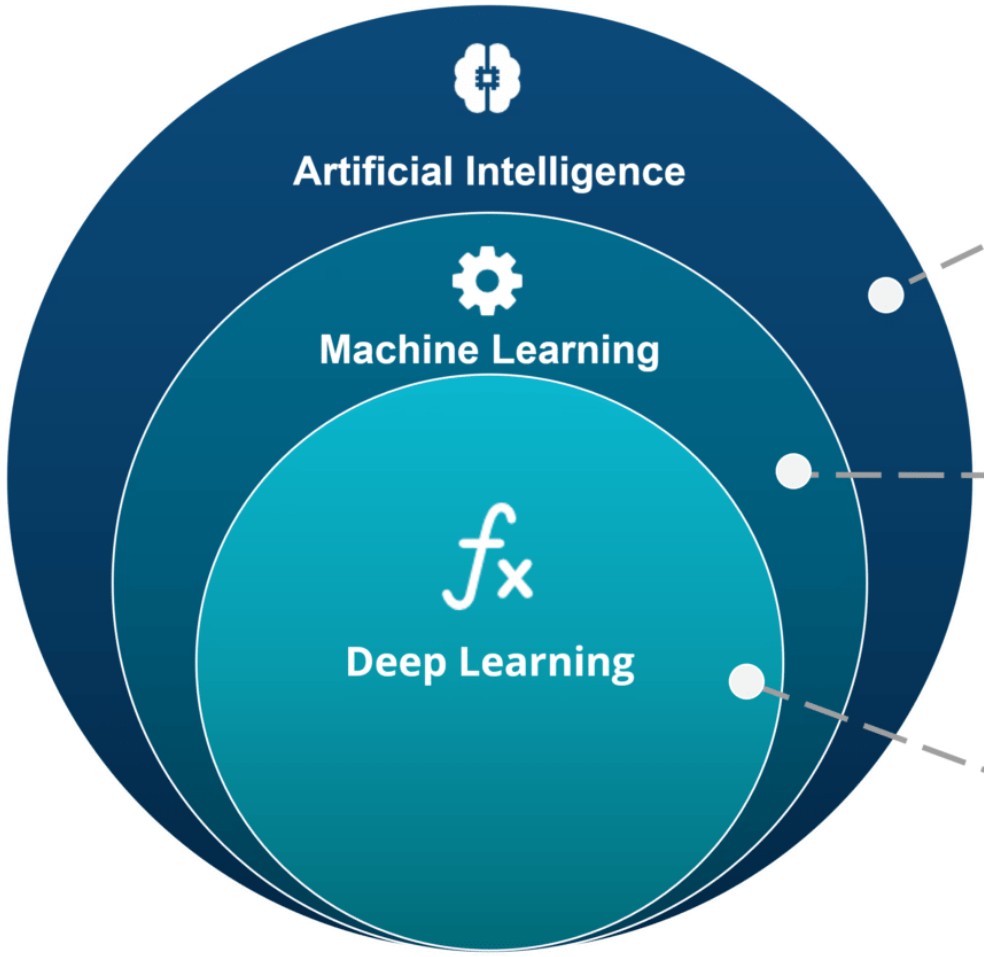
AI vs ML vs DL



ARTIFICIAL INTELLIGENCE

A technique which enables machines to mimic human behaviour

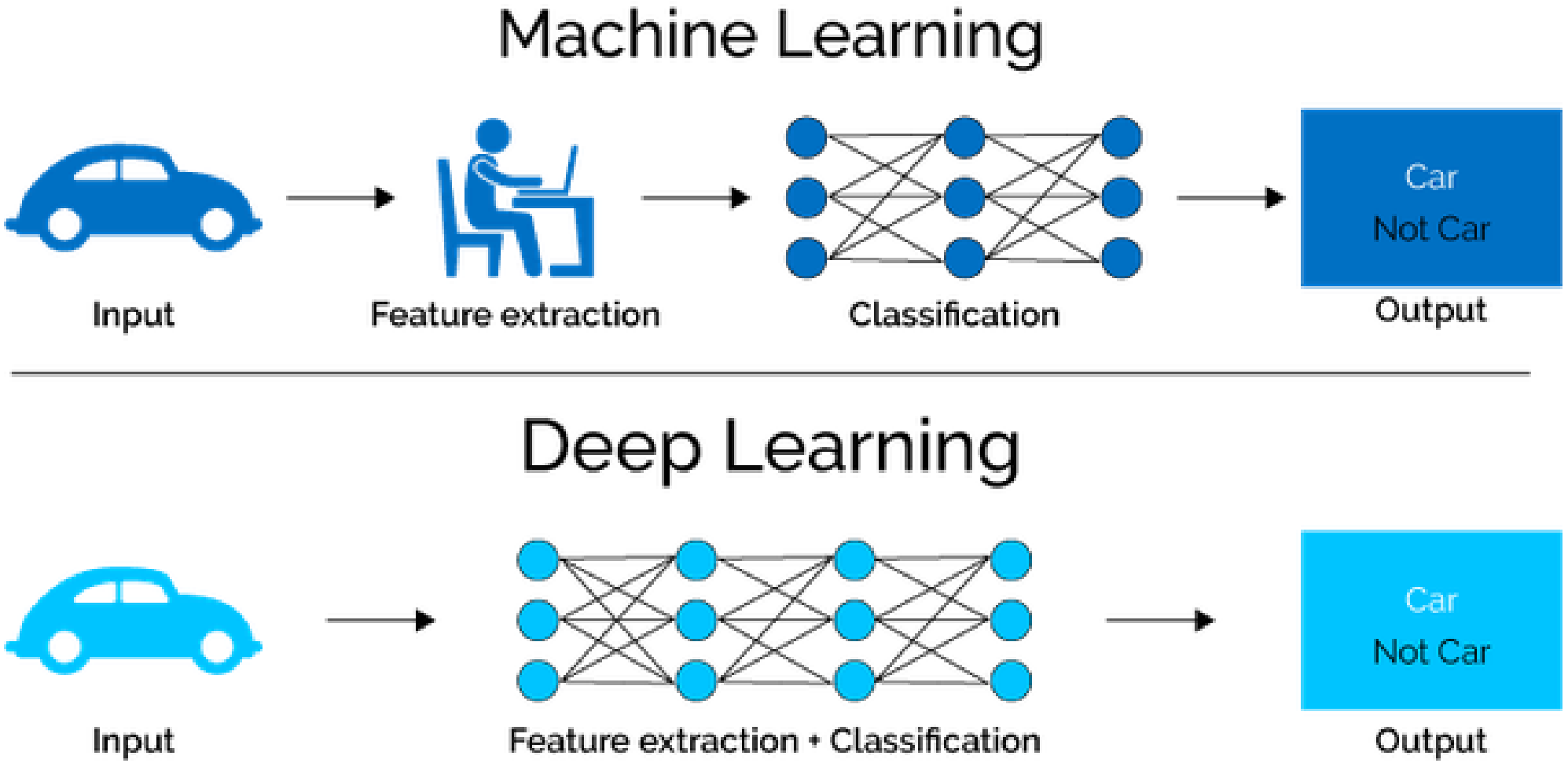
MACHINE LEARNING

Subset of AI technique which use statistical methods to enable machines to improve with experience

DEEP LEARNING

Subset of ML which make the computation of multi-layer neural network feasible

ML vs DL


Machine Learning
Input
Feature extraction
Classification
Car
Not Car
Output
Deep Learning
Input
Feature extraction + Classification
Car
Not Car
Output

What is Deep Learning

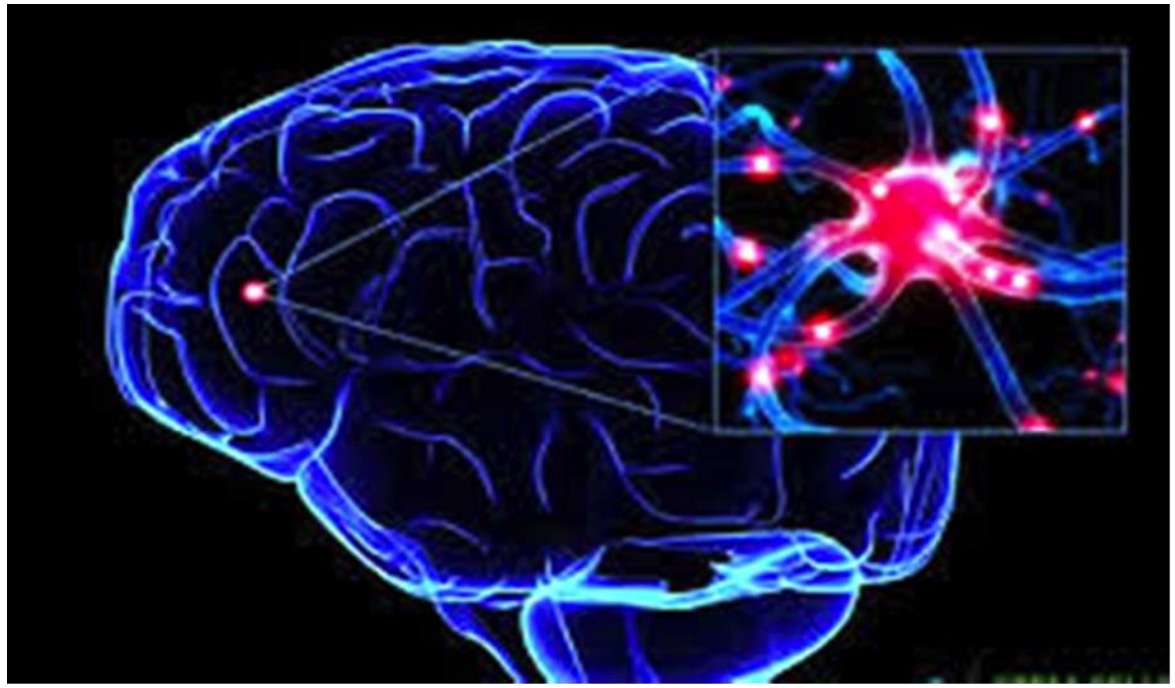
Deep learning is a machine learning technique that learns features and task directly from the data, where data may by images, text or sound!

Course Outline

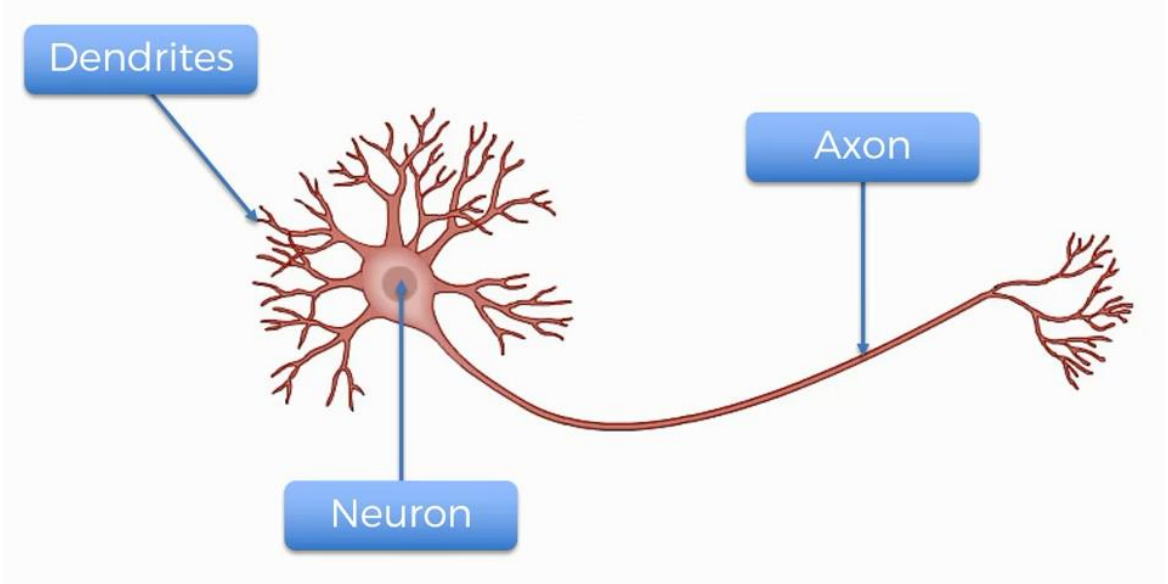
- Artificial Neural Network
- Convolutional neural network
- Recurrent neural network
- Boltzmann machine Vs Deep BM
- Self organizing maps
- Autoencoder



What is Neuron?

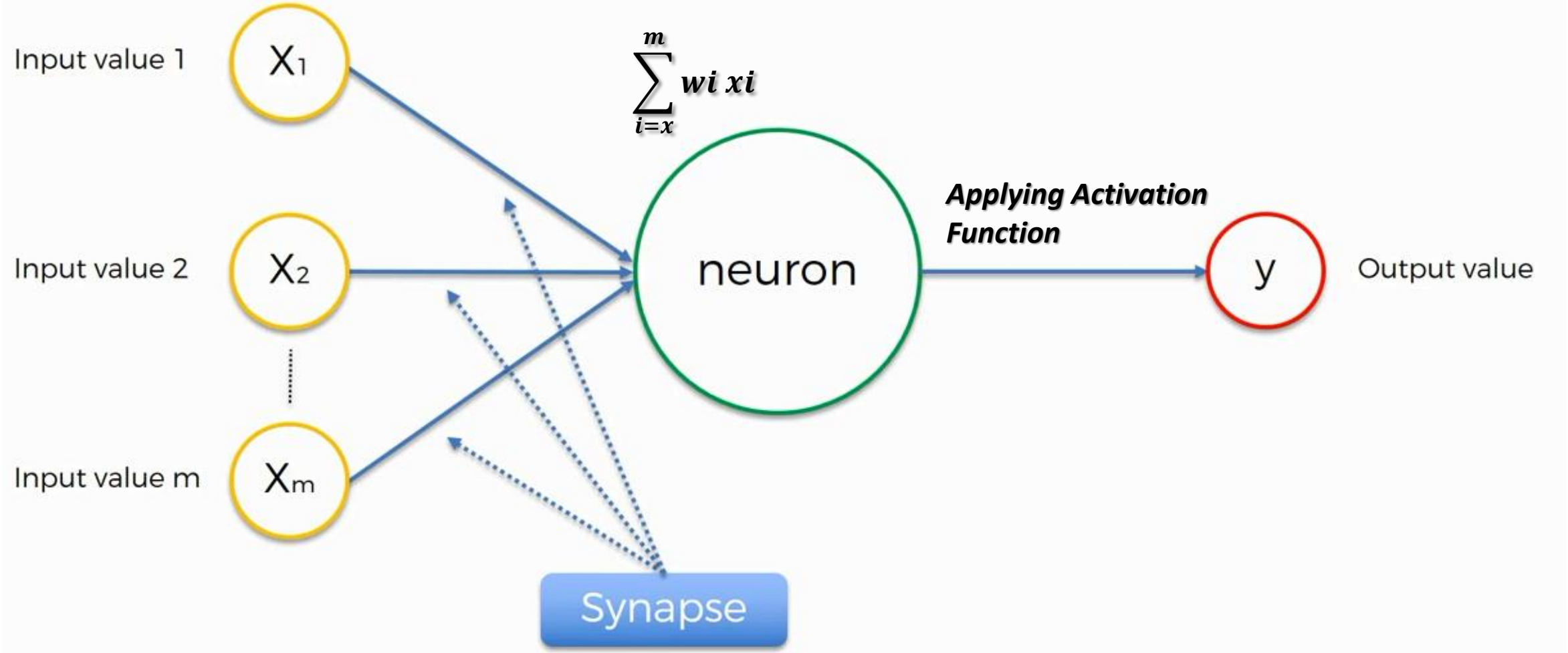


Neuron Working



Artificial Neural Network

Normalize/Standardize



What is an Activation Function?

Activation functions are an extremely important feature of the artificial neural networks. They basically decide whether a neuron should be activated or not. Whether the information that the neuron is receiving is relevant for the given information or should it be ignored.

$$Y = \text{Activation}(\Sigma(weight * input) + bias)$$

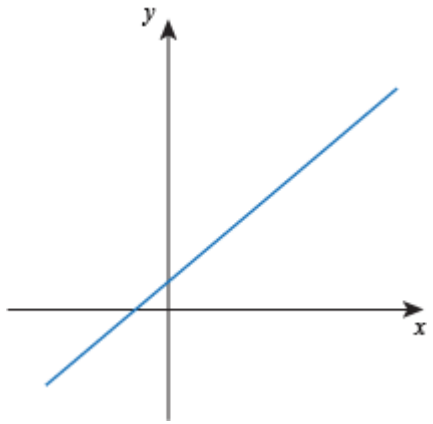
The activation function is the nonlinear transformation that we do over the input signal. This transformed output is then seen to the next layer of neurons as input.

- Linear Activation Function
- Non-Linear Activation Function

What is an Activation Function?

Linear Function

The function is a line or linear. Therefore, the output of the functions will not be confined between any range



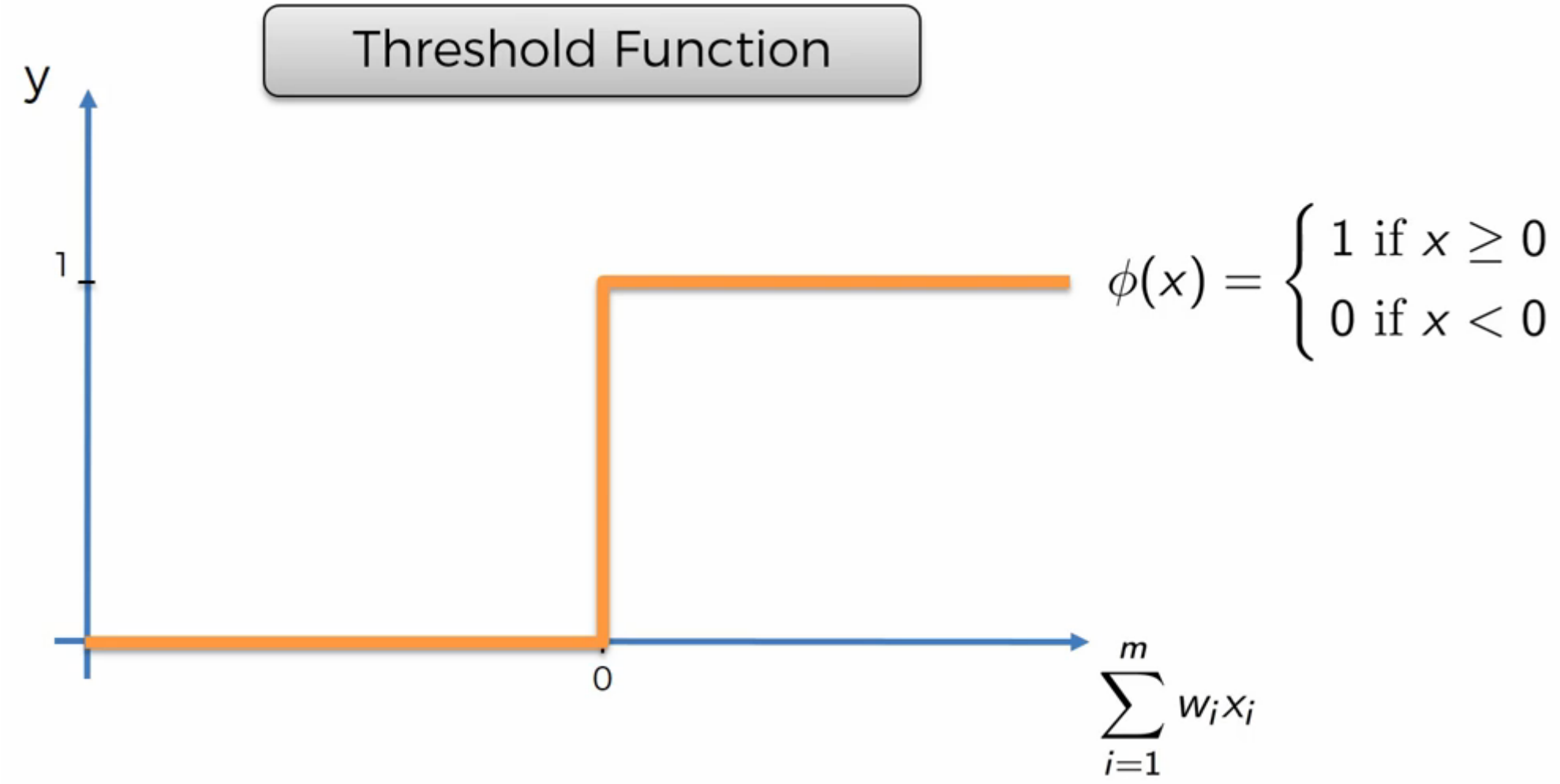
Non-Linear Function

They make it easy for the model to generalize or adapt with variety of data and to differentiate between the output

The Nonlinear Activation Functions are mainly divided on the basis of their **range or curves**

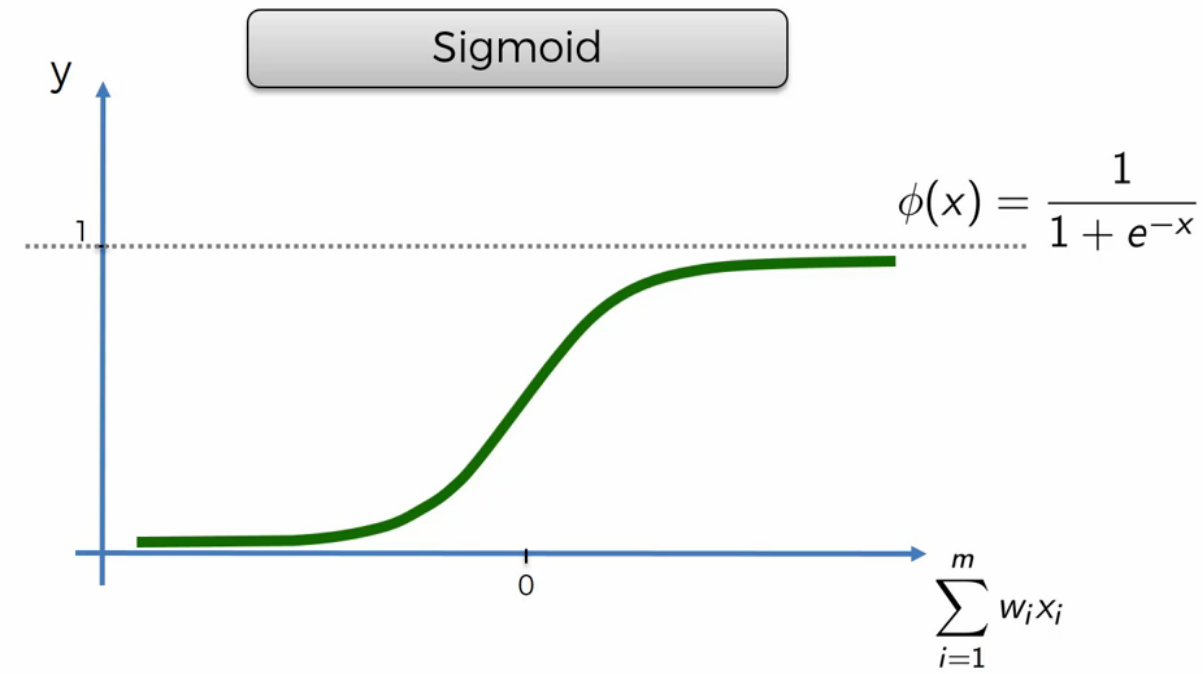
1. Threshold
2. Sigmoid
3. Tanh
4. ReLU
5. Leaky ReLU
6. Softmax

Threshold Function?



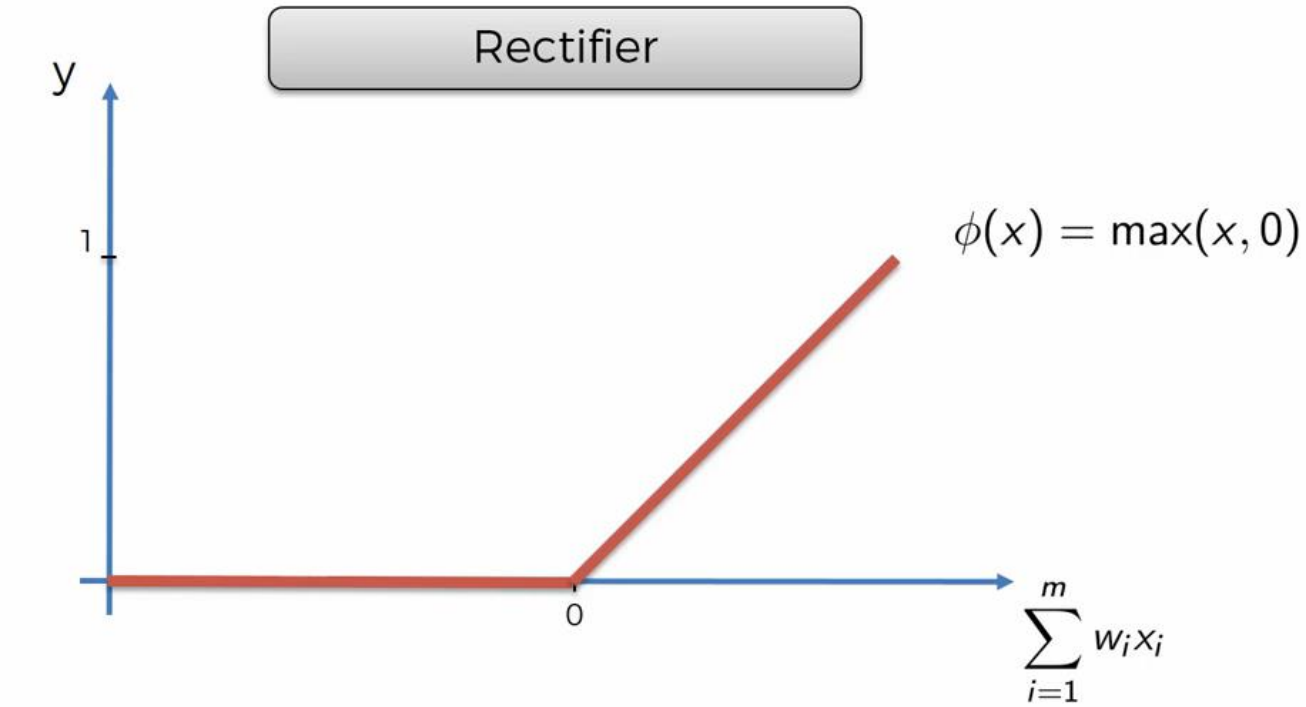
Sigmoid Function?

The Sigmoid Function curve looks like a S-shape
This function reduces extreme values or outliers in data without removing them.
It converts independent variables of near infinite range into simple probabilities between 0 and 1, and most of its output will be very close to 0 or 1.



Rectifier (ReLU) Function?

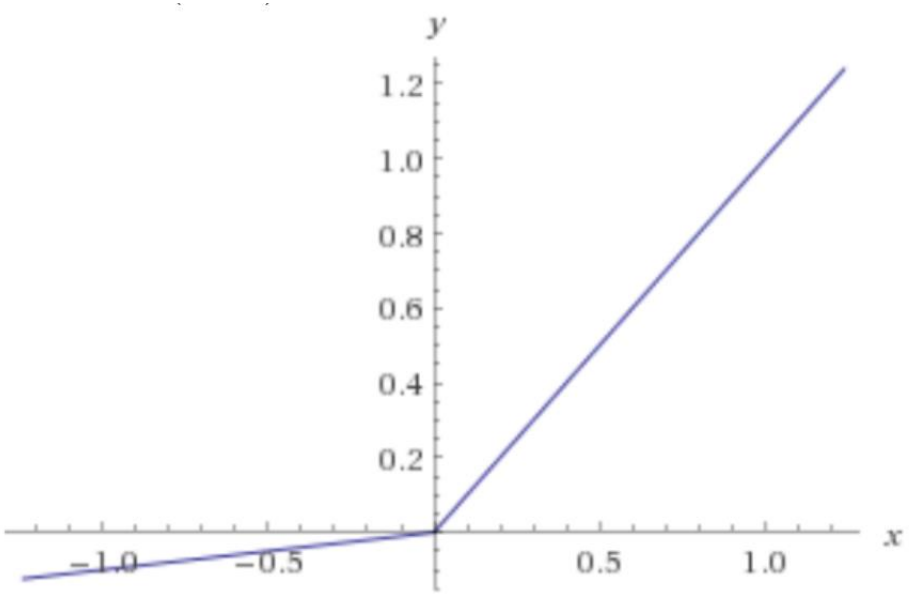
ReLU is the most widely used activation function while designing networks today. First things first, the ReLU function is nonlinear, which means we can easily backpropagate the errors and have multiple layers of neurons being activated by the ReLU function.



Leaky ReLU Function?

Leaky ReLU function is nothing but an improved version of the ReLU function. As we saw that for the ReLU function, the gradient is 0 for x<0, which made the neurons die for activations in that region. Leaky ReLU is defined to address this problem. Instead of defining the Relu function as 0 for x less than 0, we define it as a small linear component of x.

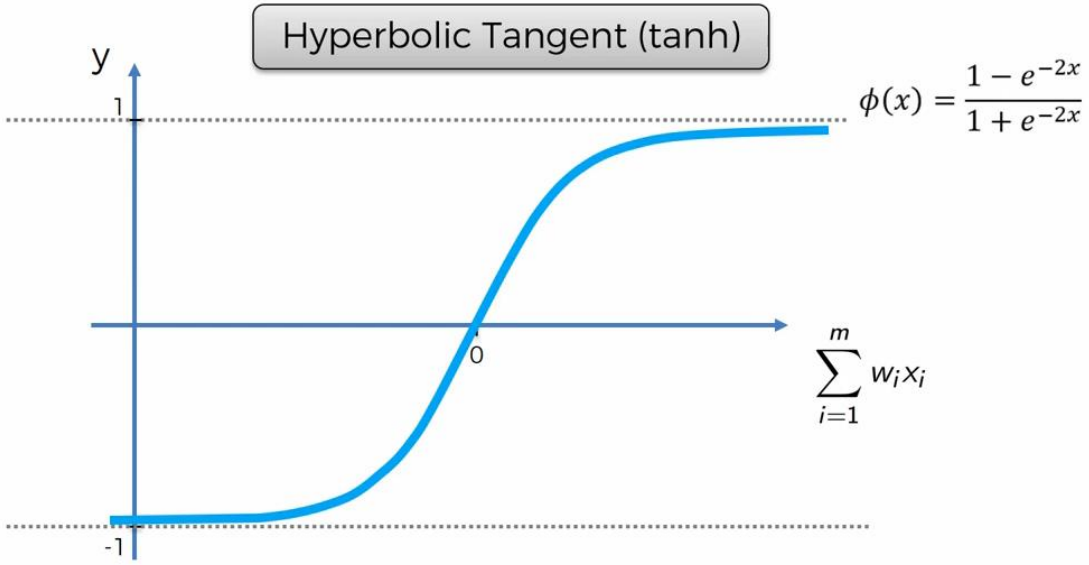
What we have done here is that we have simply replaced the horizontal line with a non-zero, non-horizontal line. Here a is a small value like 0.01 or so.



Hyperbolic Tanh

The tangent represents a ratio between the opposite and adjacent sides of a right triangle, tanh represents the ratio of the hyperbolic sine to the hyperbolic cosine: tanh(x) = sinh(x) / cosh(x)

Unlike the Sigmoid function, the normalized range of tanh is –1 to 1 The advantage of tanh is that it can deal more easily with negative numbers

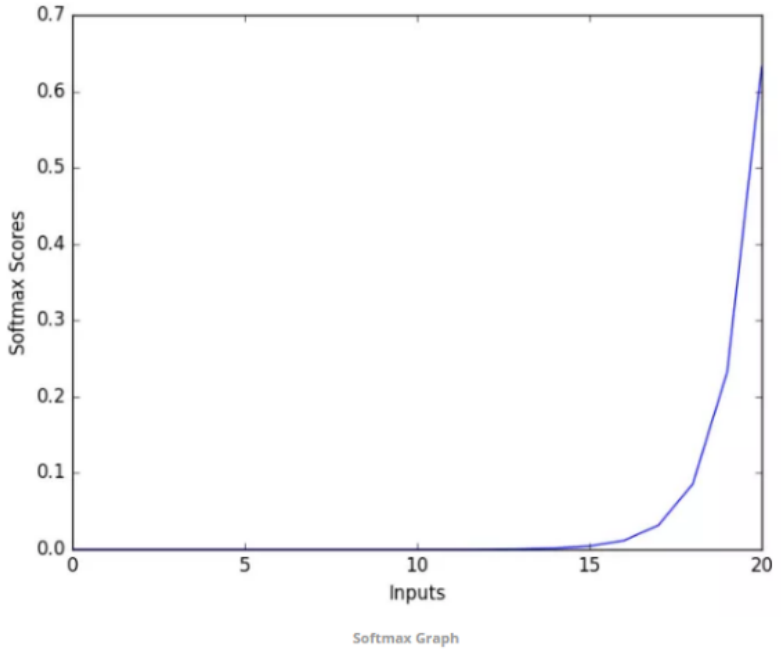


Softmax Function (for Multiple Classification)?

Softmax function calculates the probabilities distribution of the event over ‘n’ different events. In general way of saying, this function will calculate the probabilities of each target class over all possible target classes. Later the calculated probabilities will be helpful for determining the target class for the given inputs.

The main advantage of using Softmax is the output probabilities range. The range will 0 to 1, and the sum of all the probabilities will be equal to one. If the softmax function used for multi-classification model it returns the probabilities of each class and the target class will have the high probability.

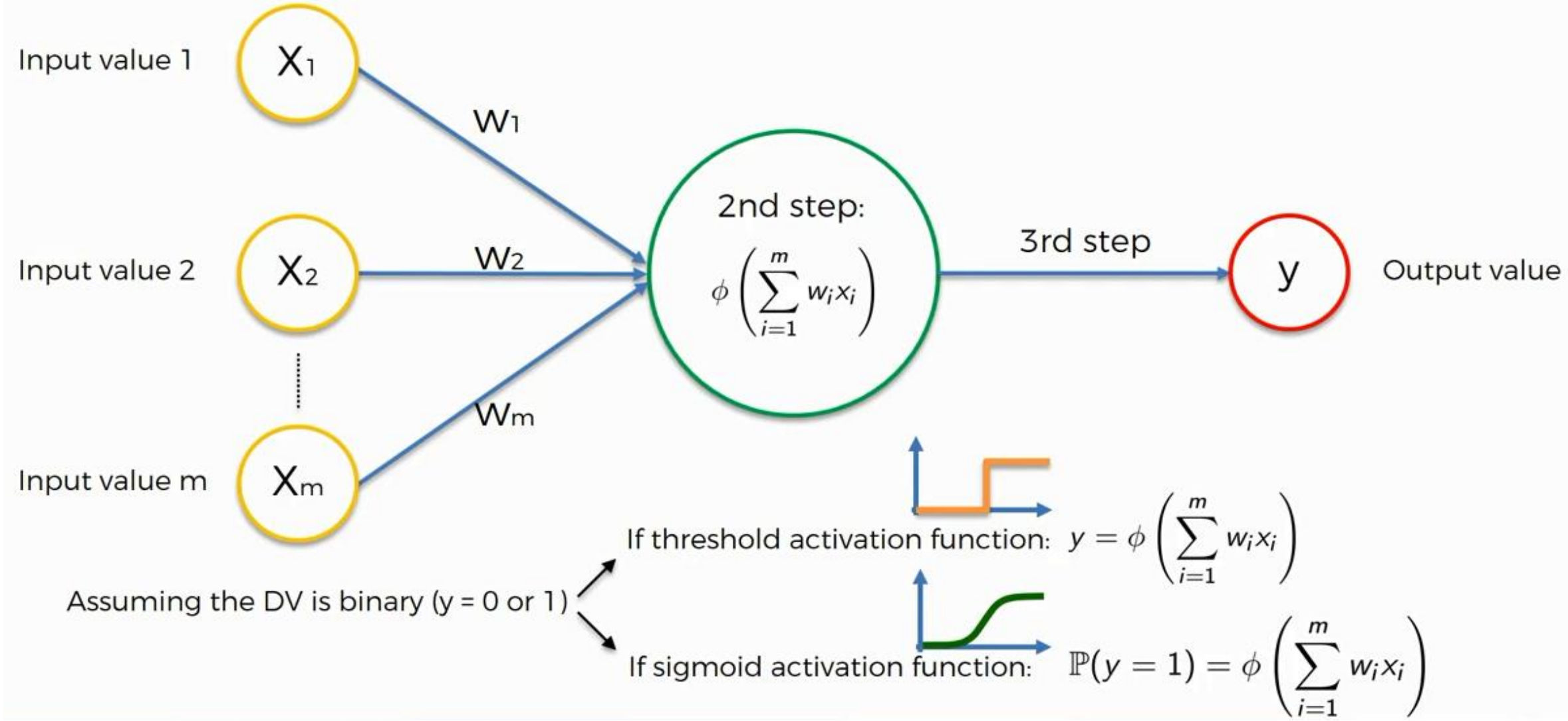
The formula computes the exponential (e-power) of the given input value and the sum of exponential values of all the values in the inputs. Then the ratio of the exponential of the input value and the sum of exponential values is the output of the softmax function.



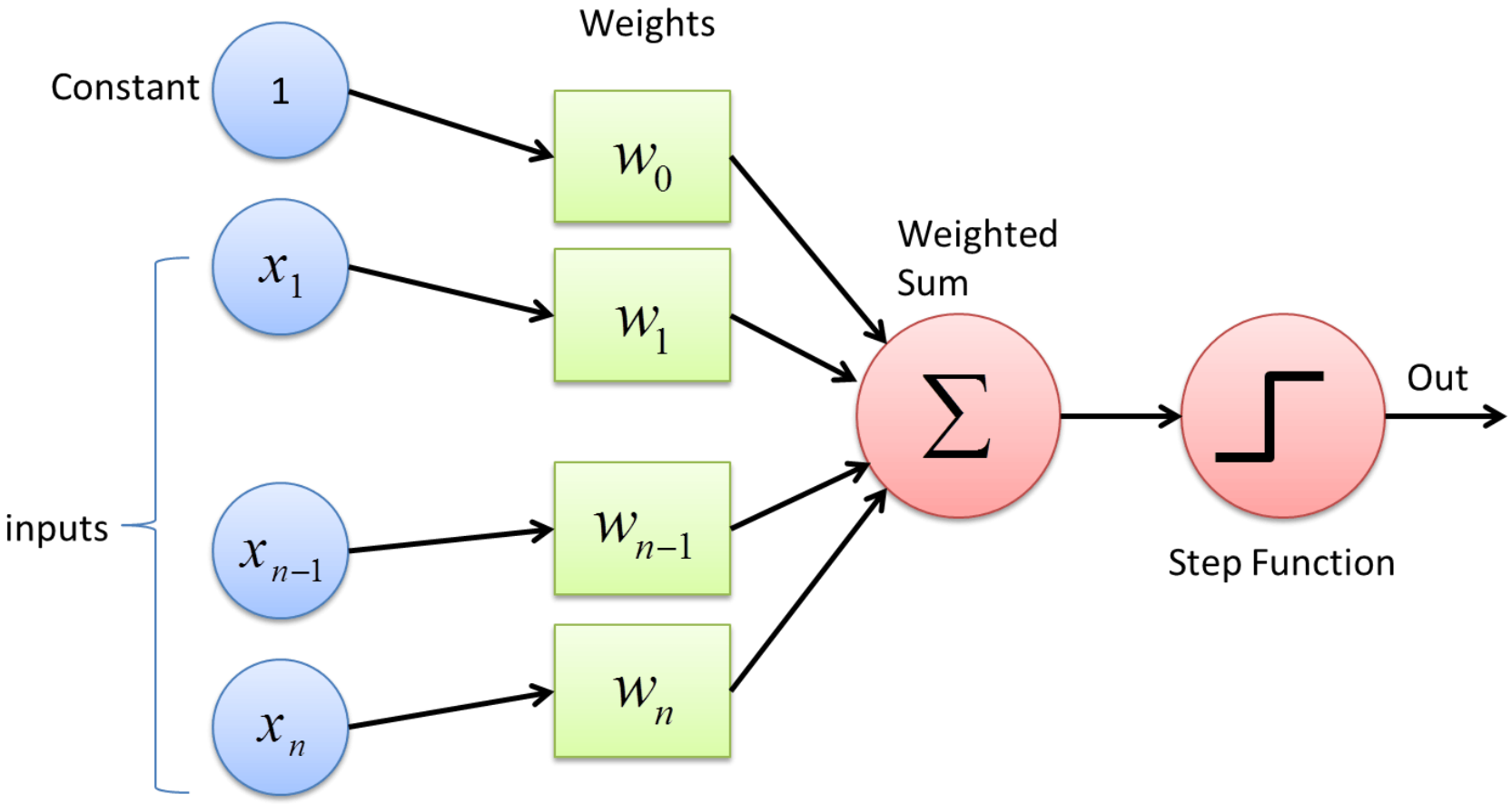
Softmax Graph

$$\text{Softmax}(x_i) = \frac{\exp(x_i)}{\sum_j \exp(x_j)}$$

Activation Function Example



Feed Forward Neural Network & Back Propagation



Hidden Layers

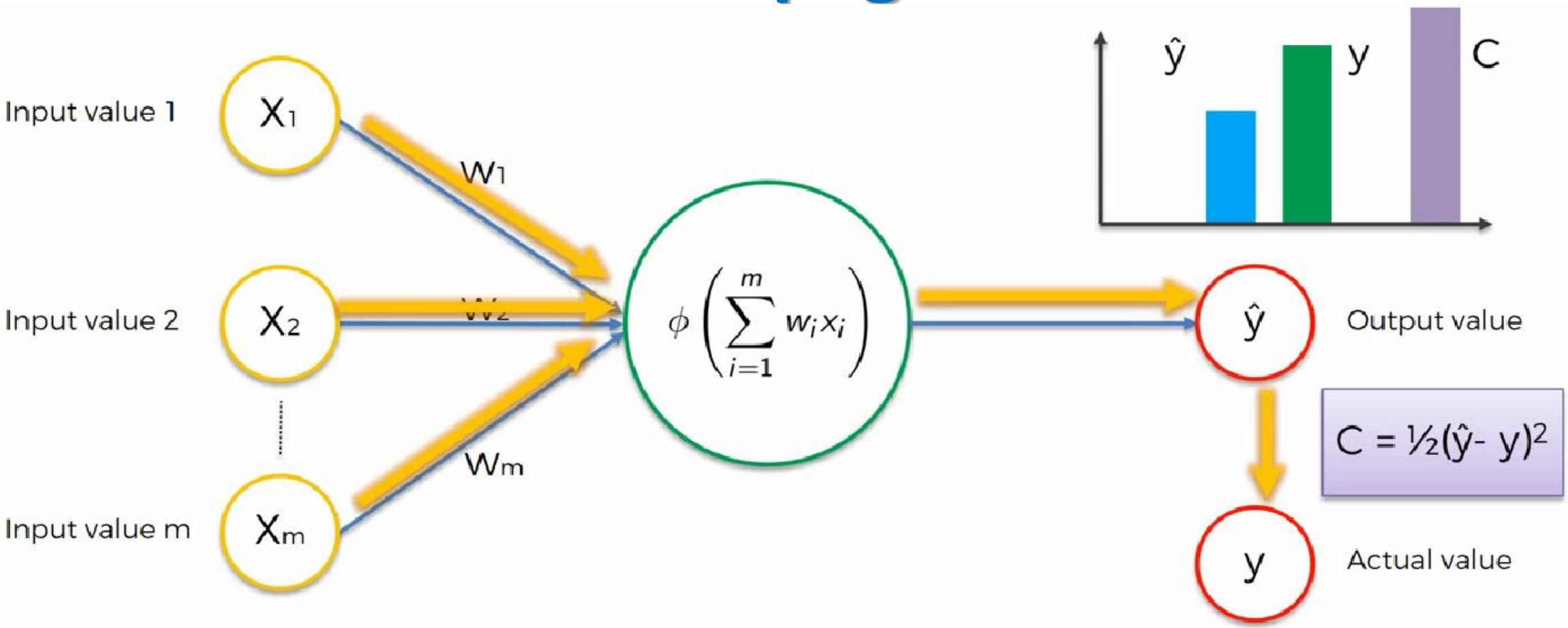


Back Propagation

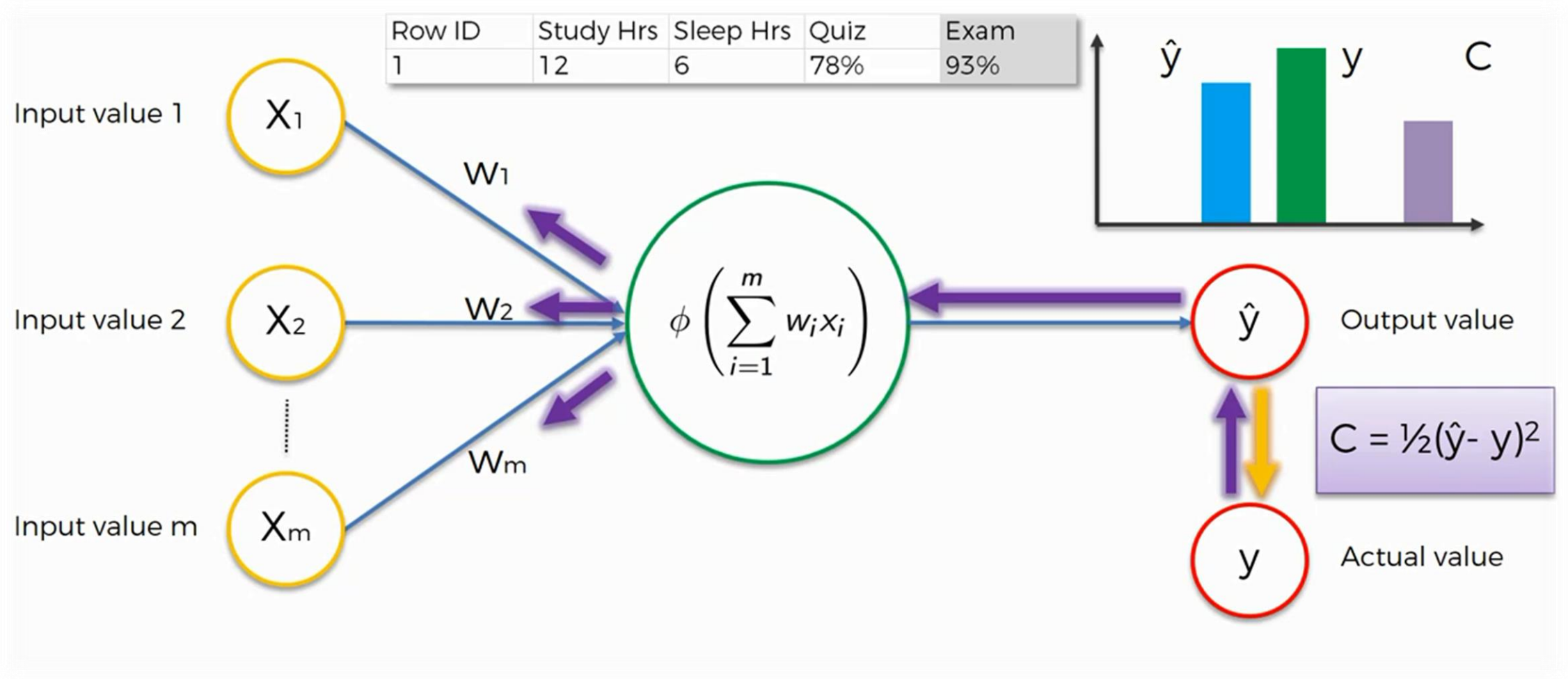
Back-propagation is the essence of neural net training. It is the method of fine-tuning the weights of a neural net based on the error rate obtained in the previous epoch (i.e., iteration). Proper tuning of the weights allows you to reduce error rates and to make the model reliable by increasing its generalization.

Backpropagation is a short form for "backward propagation of errors." It is a standard method of training artificial neural networks. This method helps to calculate the gradient of a loss function with respects to all the weights in the network.

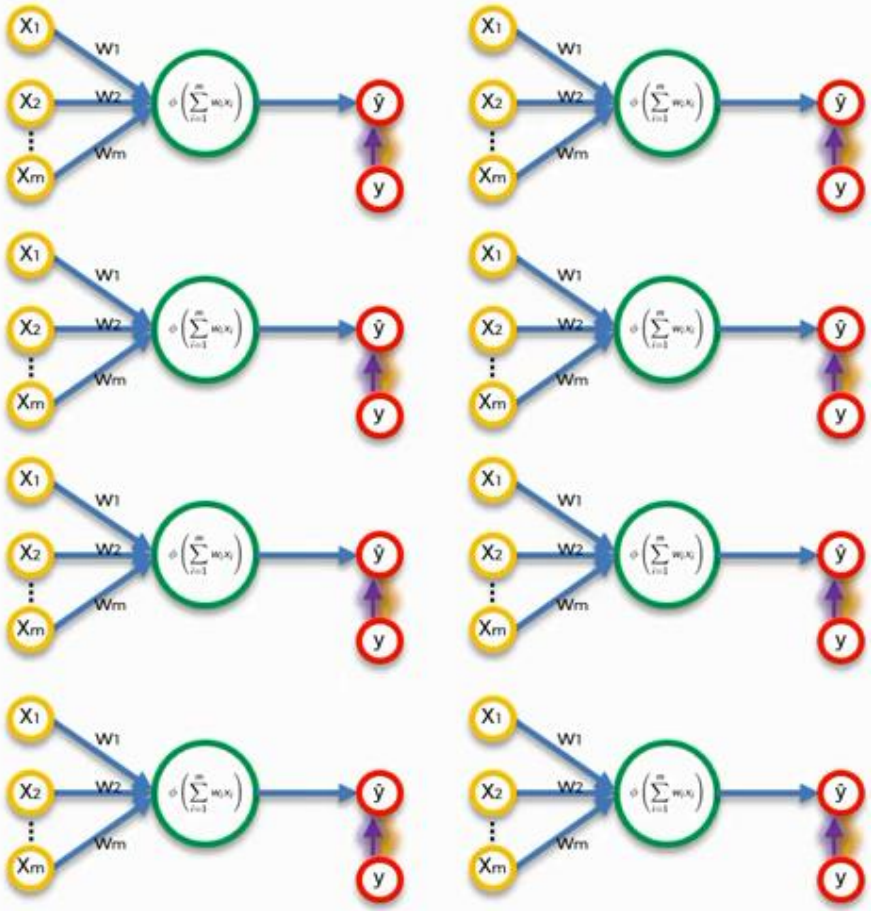
Back Propagation



Back Propagation

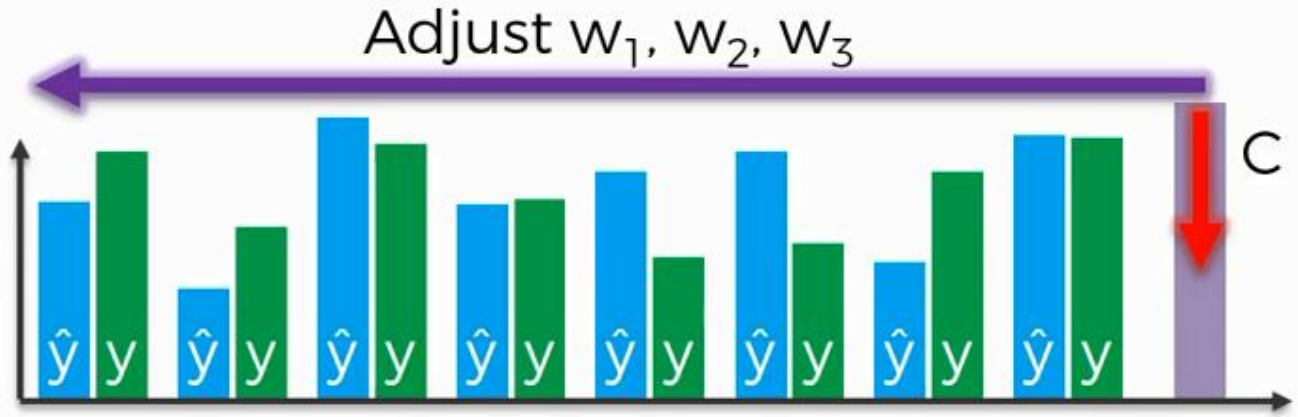


Back Propagation



Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
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2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

$$C = \sum \tfrac{1}{2}(\hat{y} - y)^2$$



What is Bias

Bias is just like an intercept added in a linear equation. It is an additional parameter in the Neural Network which is used to adjust the output along with the weighted sum of the inputs to the neuron. Moreover, bias value allows you to shift the activation function to either right or left.

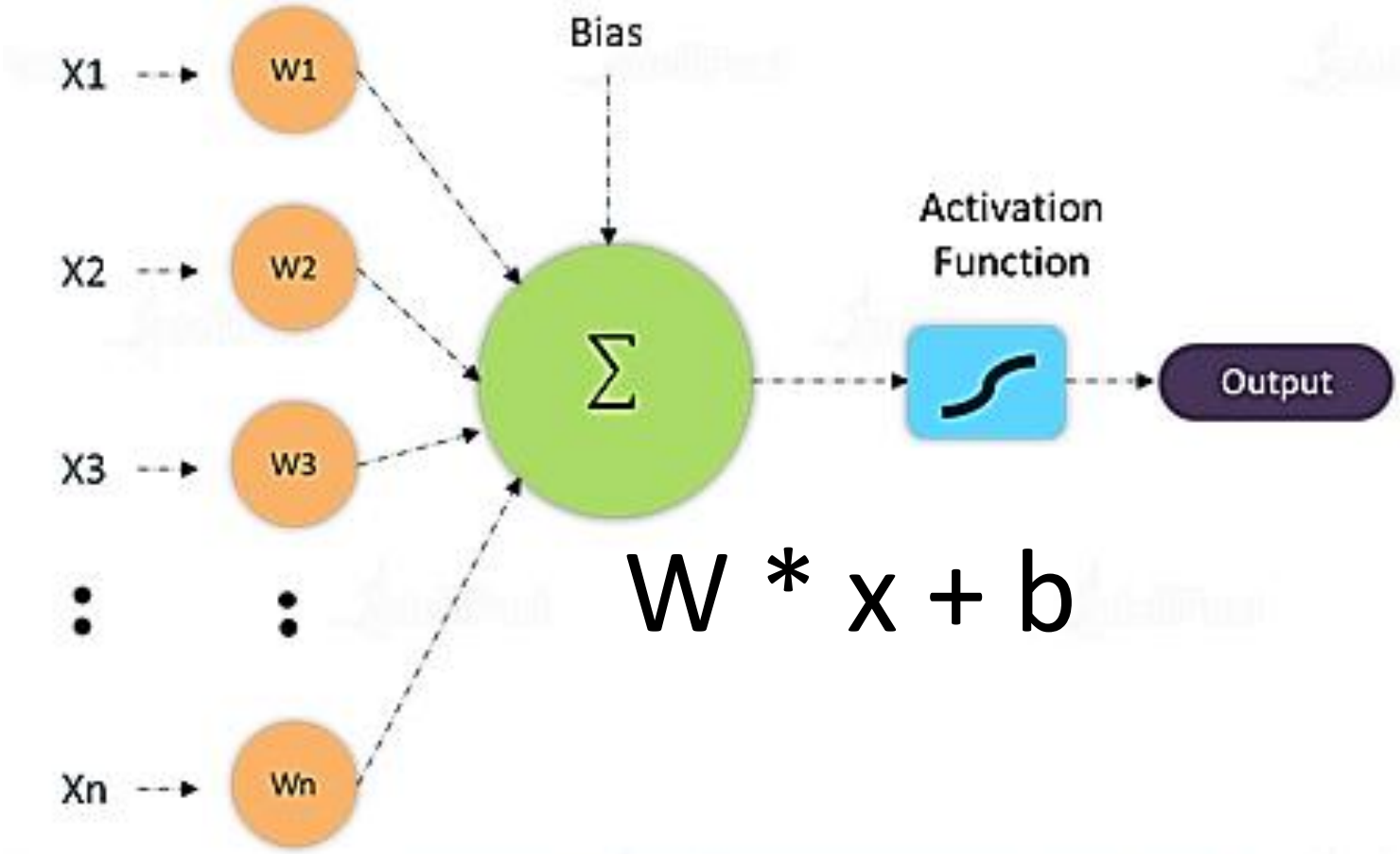
$$output = sum(weights * inputs) + bias$$

The output is calculated by multiplying the inputs with their weights and then passing it through an activation function like the Sigmoid function, etc. Here, bias acts like a constant which helps the model to fit the given data. The steepness of the Sigmoid depends on the weight of the inputs.

A simpler way to understand bias is through a constant c of a linear function

$$y = mx + c$$

What is Bias



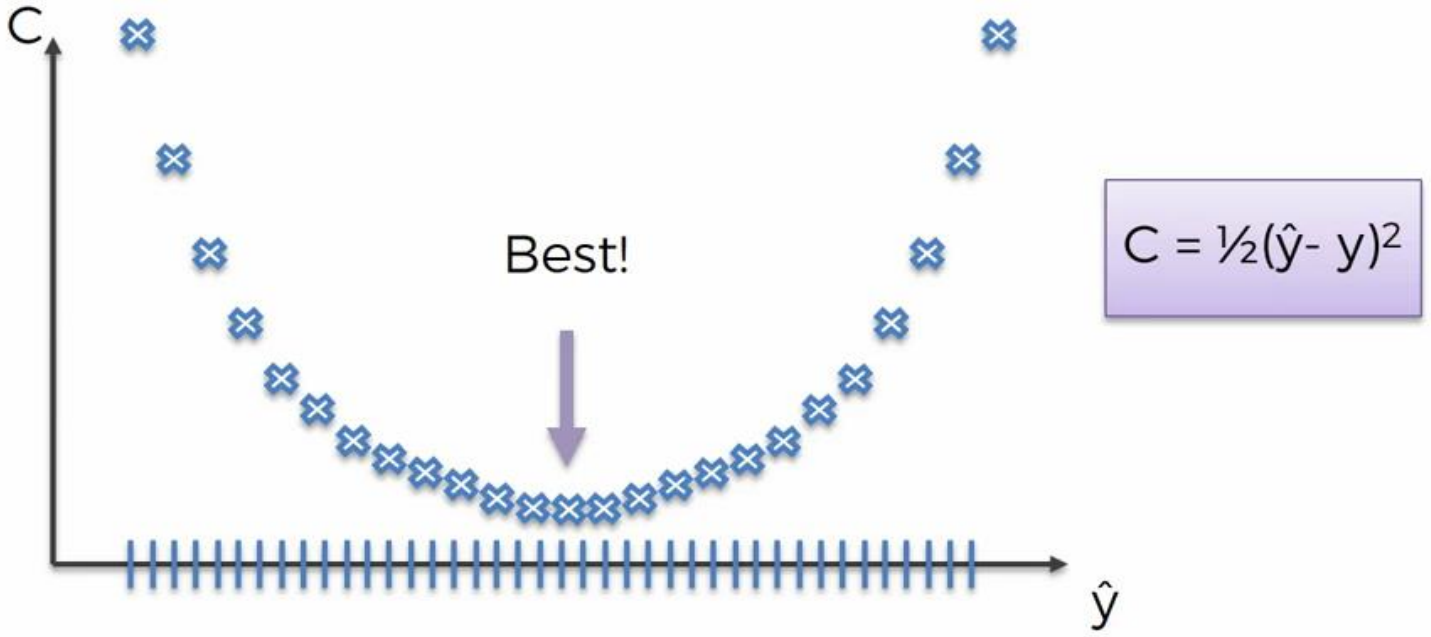
Schematic Representation of a Neuron in a Neural Network

What is Gradient Descent (BGD)

Gradient Descent is an optimization technique that is used to improve deep learning and neural network-based models by minimizing the cost function. Gradient Descent is a process that occurs in the **backpropagation** phase where the goal is to continuously resample the gradient of the model's parameter in the opposite direction based on the weight *w*, updating consistently until we reach the **global minimum** of function *J(w).*

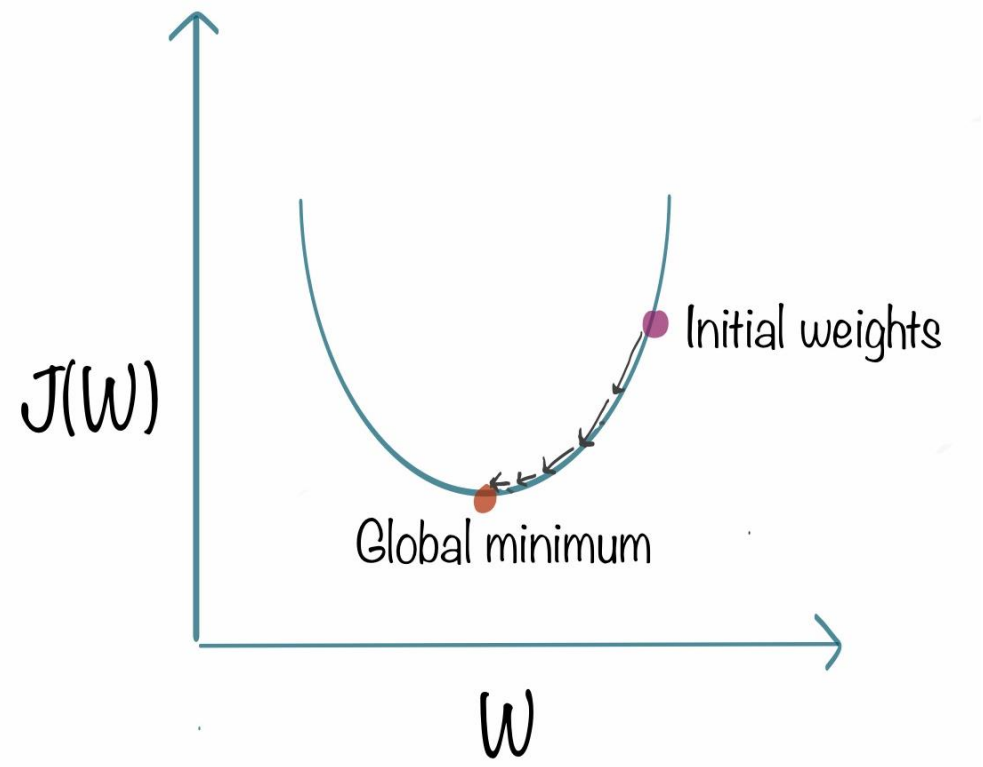
More Precisely, Gradient descent is an algorithm, which is used to iterate through different combinations of weights in an optimal way.....to find the best combination of weights which has a minimum error.

Brute force algorithm

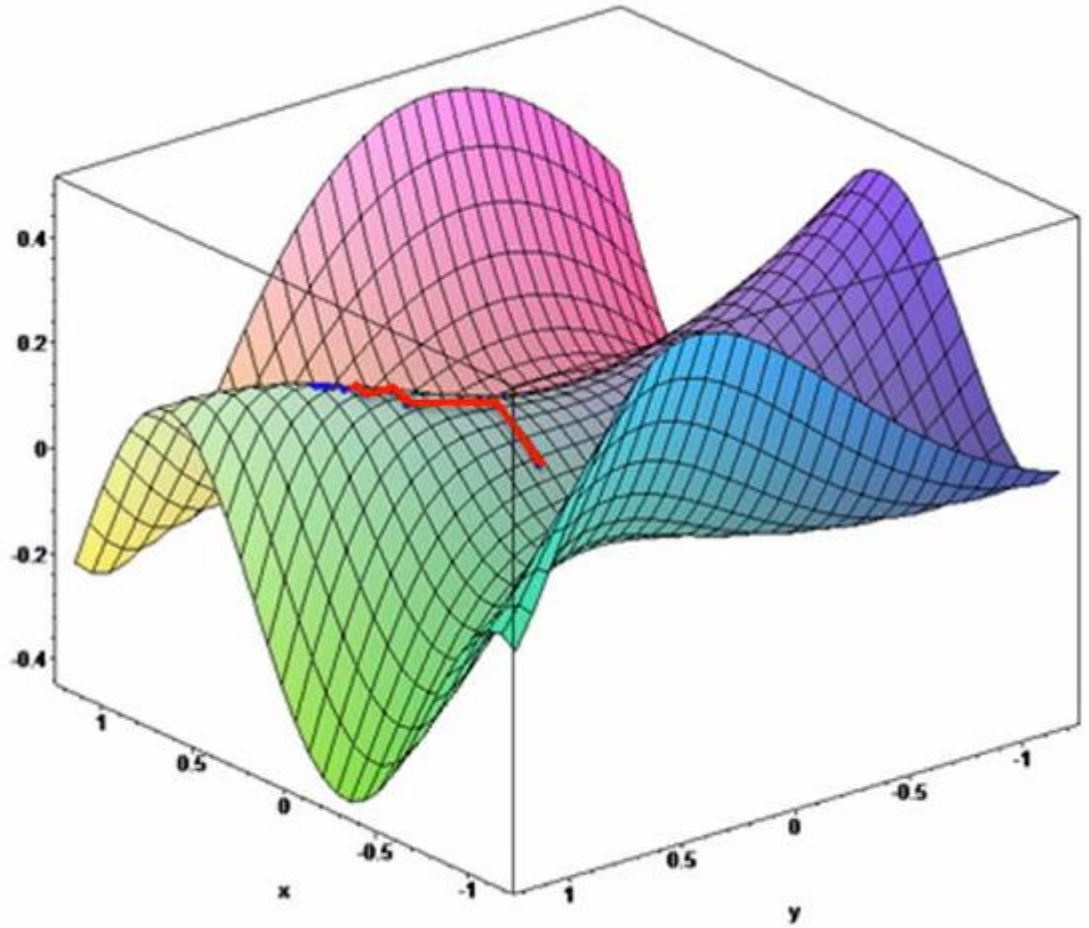


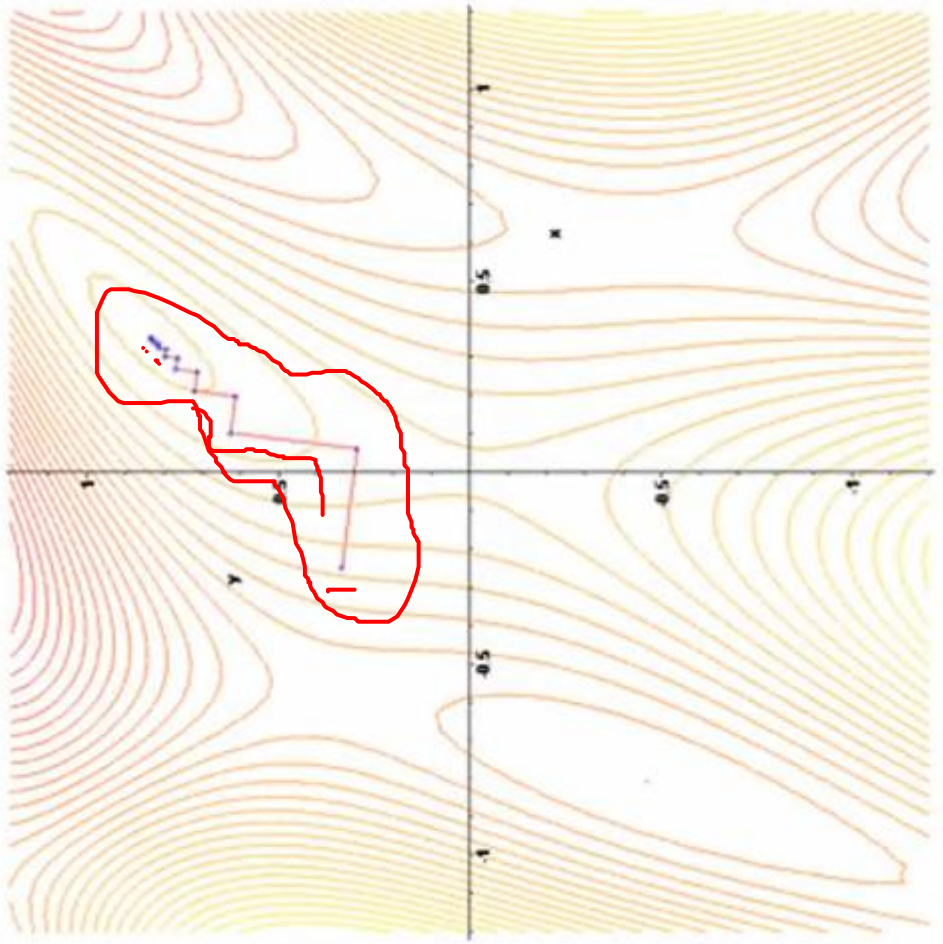
Brute Force Algorithms refers to a programming style that does not include any shortcuts to improve performance, but instead relies on sheer computing power to try all possibilities until the solution to a problem is found. A classic example is the traveling salesman problem (TSP).

What is Gradient Descent



What is Gradient Descent



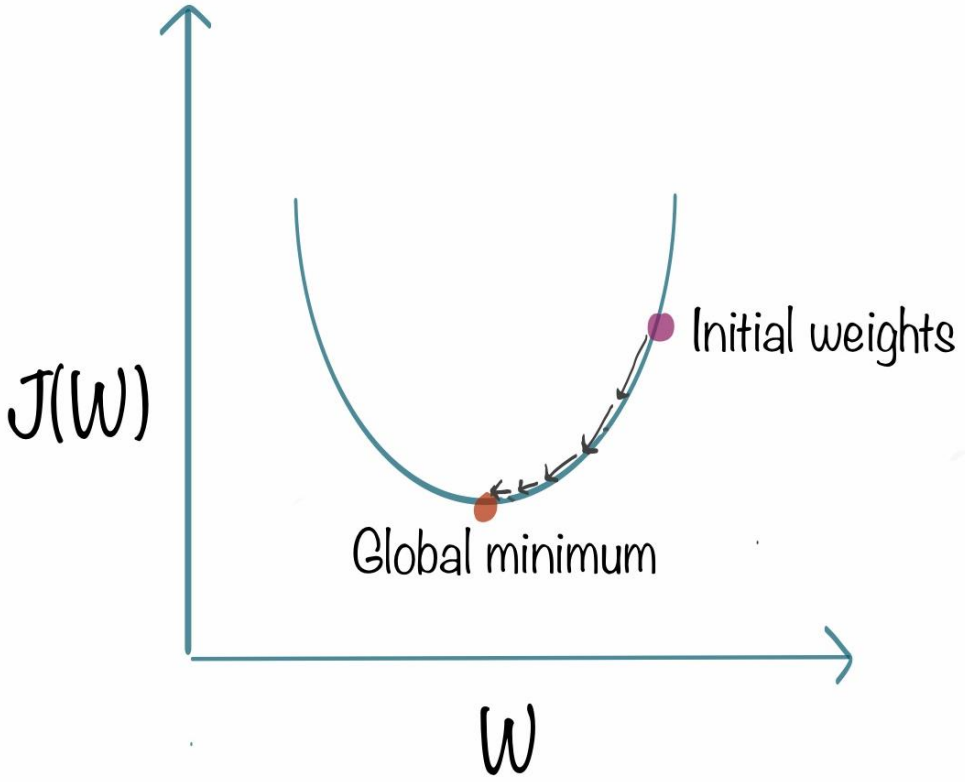


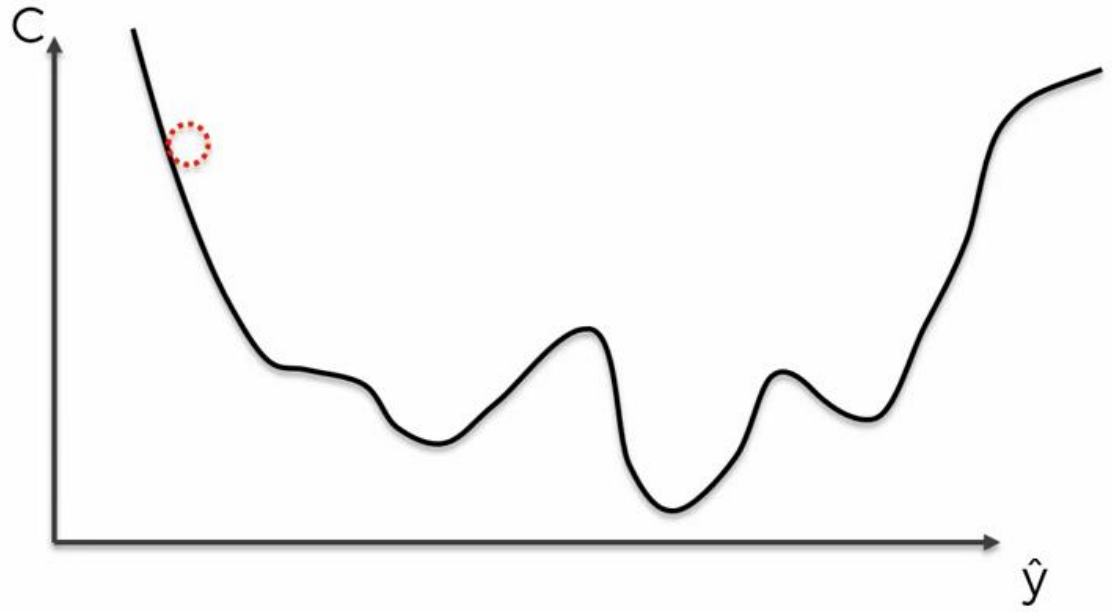
Stochastic Gradient Descent

The word '*stochastic*' means a system or a process that is linked with a random probability. Hence, in Stochastic Gradient Descent, a few samples are selected randomly instead of the whole data set for each iteration. In Gradient Descent, there is a term called "batch" which denotes the total number of samples from a dataset that is used for calculating the gradient for each iteration. In typical Gradient Descent optimization, like Batch Gradient Descent, the batch is taken to be the whole dataset. Although, using the whole dataset is really useful for getting to the minima in a less noisy or less random manner, but the problem arises when our datasets get really huge.

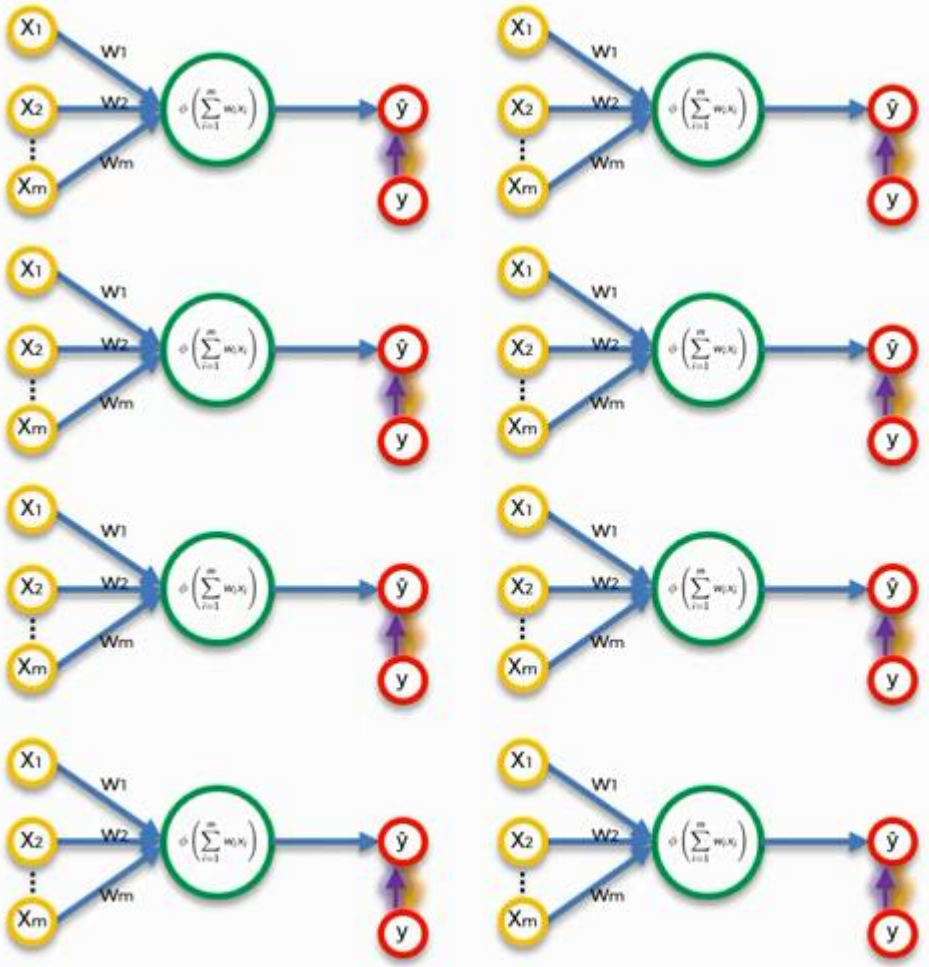
Stochastic Gradient Descent

Stochastic gradient descent (often abbreviated SGD) is an iterative method for optimizing an objective function with suitable smoothness properties (e.g., differentiable or subdifferentiable).



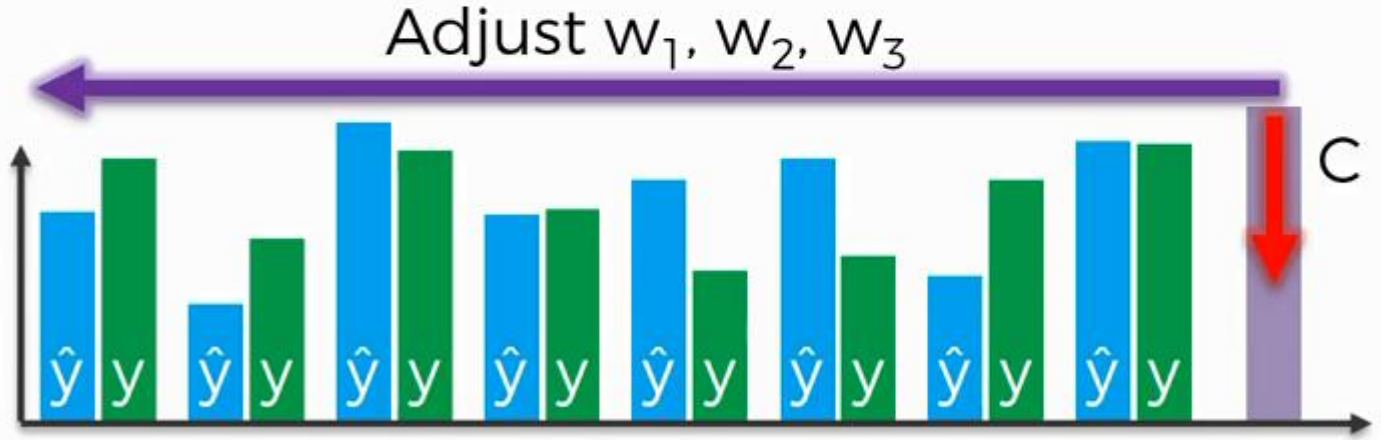


Stochastic Gradient Descent



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3	115	4	100%	95%
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$$C = \sum \tfrac{1}{2}(\hat{y} - y)^2$$



Stochastic gradient descent

Upd w's

Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
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6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

Batch Gradient Descent

Upd w's
Upd w's
Upd w's
Upd w's
Upd w's
Upd w's
Upd w's
Upd w's

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5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

Stochastic Gradient Descent

Mini Batch gradient descent

Mini-batch gradient descent is a variation of the gradient descent algorithm that splits the training dataset into small batches that are used to calculate model error and update model coefficients.
Implementations may choose to sum the gradient over the mini-batch which further reduces the variance of the gradient.

Mini-batch gradient descent seeks to find a balance between the robustness of stochastic gradient descent and the efficiency of batch gradient descent. It is the most common implementation of gradient descent used in the field of deep learning.

Mini Batch gradient descent

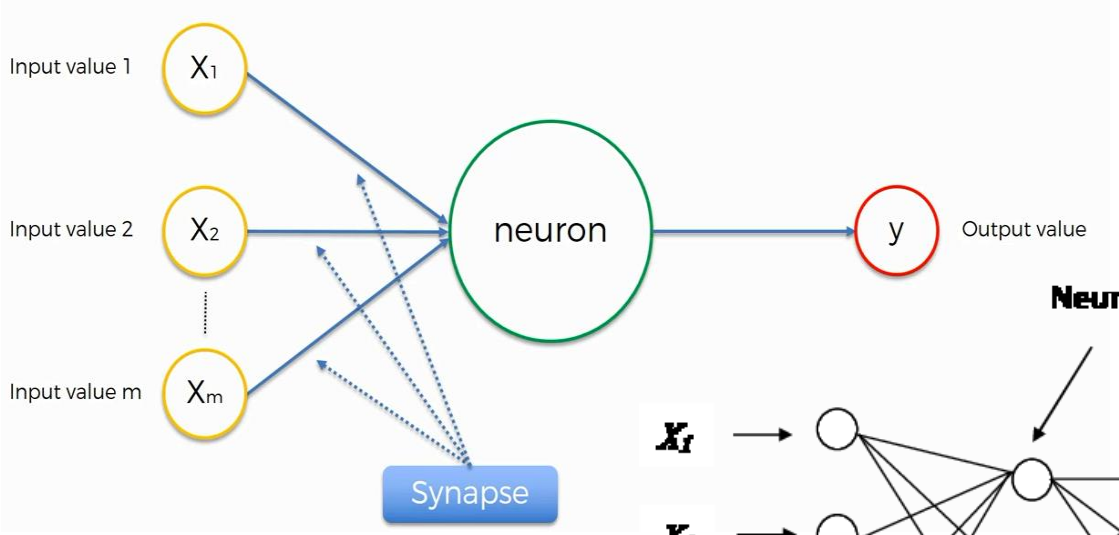
car_ID	symboling	CarName	fueltype	aspiration	doornumber	carbody	drivewheel	enginelocatic	wheelbase	carlength	carwidth	carheight	curbweight
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2	3	alfa-romero	gas	std	two	convertible	rwd	front	88.6	168.8	64.1	48.8	2548
3	1	alfa-romero	gas	std	two	hatchback	rwd	front	94.5	171.2	65.5	52.4	2823
4	2	audi 100 ls	gas	std	four	sedan	fwd	front	99.8	176.6	66.2	54.3	2337
5	2	audi 100ls	gas	std	four	sedan	4wd	front	99.4	176.6	66.4	54.3	2824
6	2	audi fox	gas	std	two	sedan	fwd	front	99.8	177.3	66.3	53.1	2507
7	1	audi 100ls	gas	std	four	sedan	fwd	front	105.8	192.7	71.4	55.7	2844
8	1	audi 5000	gas	std	four	wagon	fwd	front	105.8	192.7	71.4	55.7	2954
9	1	audi 4000	gas	turbo	four	sedan	fwd	front	105.8	192.7	71.4	55.9	3086
10	0	audi 5000s (	gas	turbo	two	hatchback	4wd	front	99.5	178.2	67.9	52	3053
11	2	bmw 320i	gas	std	two	sedan	rwd	front	101.2	176.8	64.8	54.3	2395
12	0	bmw 320i	gas	std	four	sedan	rwd	front	101.2	176.8	64.8	54.3	2395
13	0	bmw x1	gas	std	two	sedan	rwd	front	101.2	176.8	64.8	54.3	2710
14	0	bmw x3	gas	std	four	sedan	rwd	front	101.2	176.8	64.8	54.3	2765
15	1	bmw z4	gas	std	four	sedan	rwd	front	103.5	189	66.9	55.7	3055
16	0	bmw x4	gas	std	four	sedan	rwd	front	103.5	189	66.9	55.7	3230
17	0	bmw x5	gas	std	two	sedan	rwd	front	103.5	193.8	67.9	53.7	3380
18	0	bmw x3	gas	std	four	sedan	rwd	front	110	197	70.9	56.3	3505

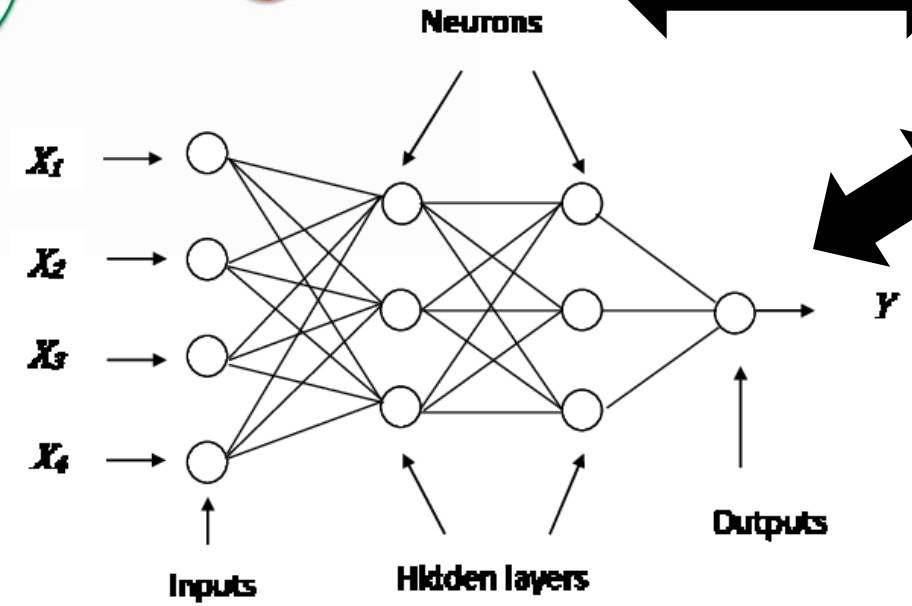
Different types of Neural Network

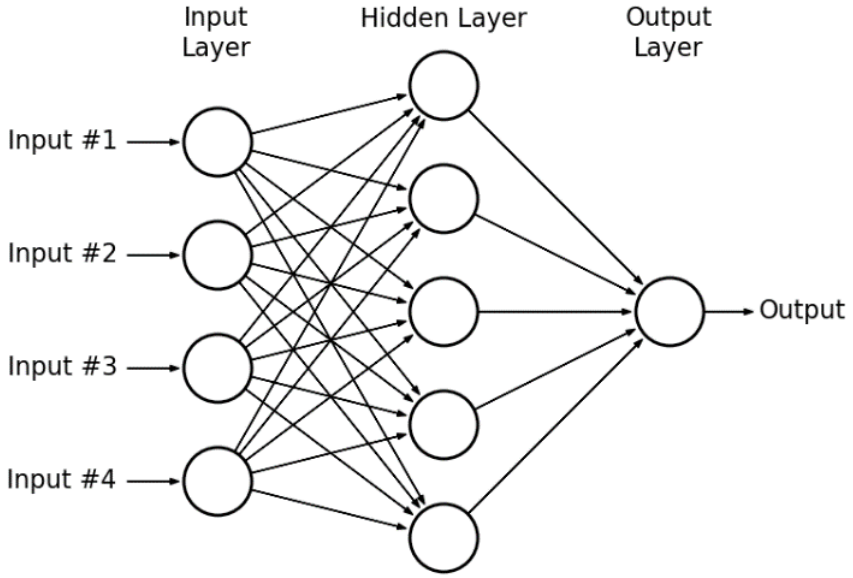
- Perceptron (Multilayer Perceptron) & ANN
- Feedforward Neural Network – Artificial Neuron
- Radial Basis Function Neural Network
- Convolutional Neural Network (CNN)
- Recurrent Neural Network(RNN)
- Long Short-Term Memory (LSTM)

Perceptron (Multilayer Perceptron)

A perceptron is a network with two layers, one input and one output. ... Artificial neural network, which has input layer, output layer, and two or more trainable weight layers (consisting of Perceptron) is called multilayer perceptron or MLP





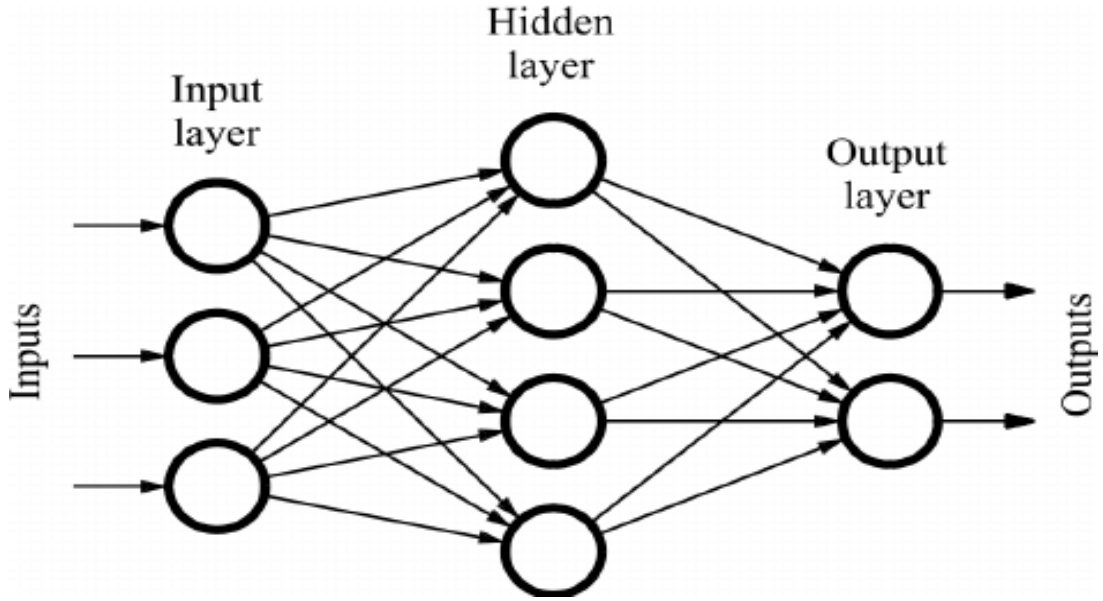


Feedforward Neural Network

It is one of the simplest types of artificial neural networks. In a feedforward neural network, the data passes through different input nodes until it reaches the output node. In other words, the data moves in only one direction from the first range until it reaches the output node. It is also known as a front propagating wave which is usually obtained using a graded activation function. Unlike more complex types of neural networks, backpropagation and data move in only one direction. A feedforward neural network may consist of a single layer or may contain hidden layers. In a feedful neural network, the products of the inputs and their weights are calculated. This is then fed to the output.

whereas
Backpropagation is a training algorithm consisting of 2 steps:

- Feedforward the values.
- Calculate the error and propagate it back to the earlier layers.



Radial Basis Function Neural Network

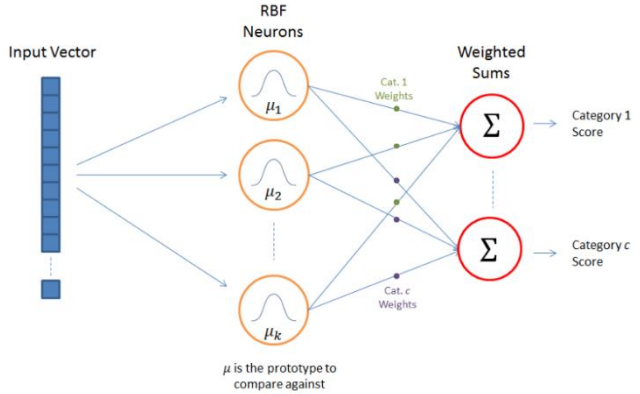
A radial basis function (RBF) is a function that assigns a real value to each input from its domain (it is a real-value function), and the value produced by the RBF is always an absolute value; i.e., it is a measure of distance and cannot be negative

$$f(x) = f(||x||)$$

Euclidean distance, the straight-line distance between two points in Euclidean space, is typically used. Radial basis functions are used to approximate functions, much as neural networks act as function approximators.
RBF network represents a radial basis function network. The radial basis functions act as activation functions.
The approximant f(x) is differentiable with respect to the weights W, which are learned using iterative updater methods coming among neural networks.

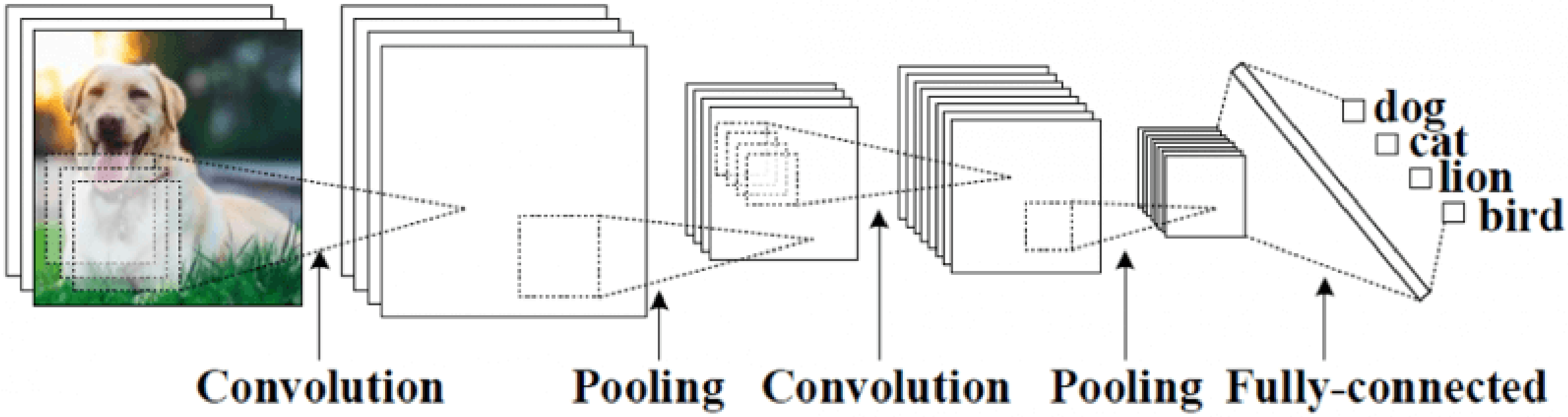
Radial basis function neural networks are extensively applied in power restoration systems. In recent decades, power systems have become larger and more complex.
This increases the risk of blackout. This neural network is used in power restoration systems to restore power in the least amount of time.





Convolutional Neural Network

Convolutional Neural Networks (CNN) is one of the variants of neural networks used heavily in the field of Computer Vision. It derives its name from the type of hidden layers it consists of. The hidden layers of a CNN typically consist of convolutional layers, pooling layers, fully connected layers, and normalization layers. Here it simply means that instead of using the normal activation functions defined above, convolution and pooling functions are used as activation functions.



Recurrent Neural Network

Recurrent Neural Network(RNN) are a type of Neural Network where the output from previous step are fed as input to the current step. In traditional neural networks, all the inputs and outputs are independent of each other, but in cases like when it is required to predict the next word of a sentence, the previous words are required and hence there is a need to remember the previous words. Thus, RNN came into existence, which solved this issue with the help of a Hidden Layer. The main and most important feature of RNN is Hidden state, which remembers some information about a sequence.

